

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458800.5866 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.023 +/- 0.058
Transit depth (Rp/Rs)^2	0.053 +/- 0.27 [%]
Orbital Inclination (inc)	84.09 +/- 1.81
Airmass coefficient 1 (a1)	1.016 +/- 0.051
Airmass coefficient 2 (a2)	-0.009 +/- 0.026
Scatter in the residuals of the lightcurve fit is	4.98 %
Best Comparison Star	#3 - [307, 57]
Optimal Aperture	4.35
Optimal Annulus	9.55
Transit Duration (day)	0.01 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.023 ± 0.058 vs 0.1714 (deviation 2.25σ)
Duration fit vs published	0.01 d vs 0.0892 d (deviation 3.86σ)
Mid-time O–C	1.6 min
Depth SNR	0.3
Residual scatter	4.98 %

Light curve

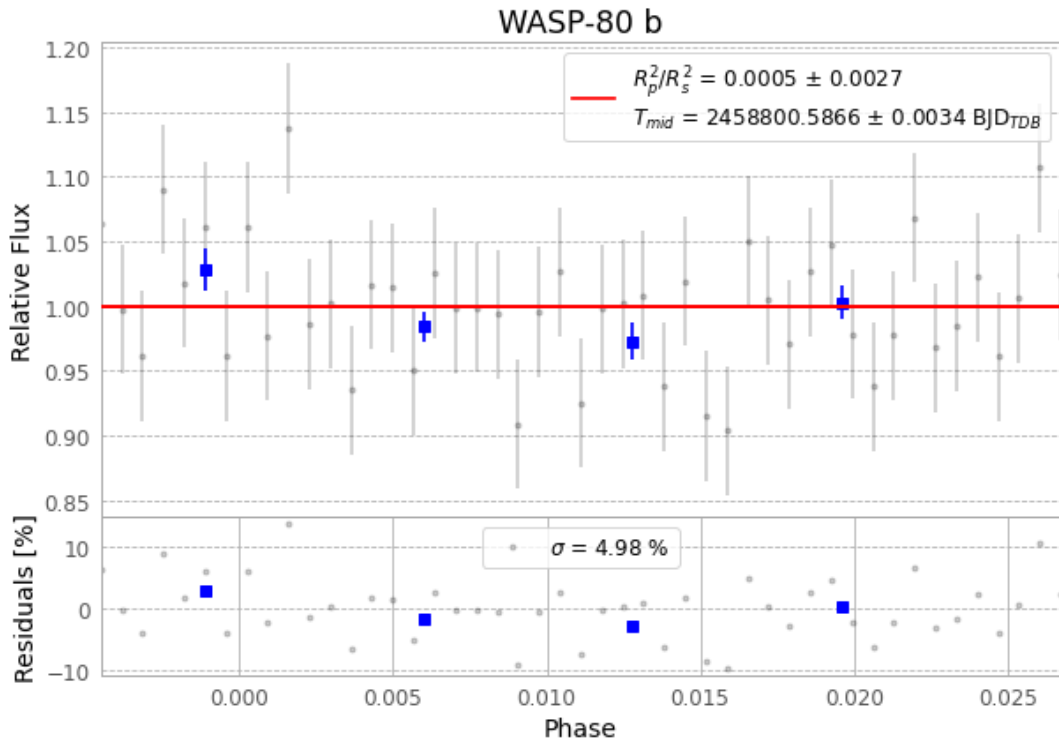


Figure 1 — FinalLightCurve_WASP-80 b_2019-11-12.png (SHA-256 4770448d0ad64638...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c5dd2321f0ebeb2f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

47 FITS frames (31 MB), WASP-80191113014512.FITS → WASP-80191113040312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-191113.log (d5f27407873f...), FinalLightCurve_WASP-80 b_2019-11-12.png (4770448d0ad6...), FinalLightCurve_WASP-80 b_2019-11-12.csv (b0491202ba63...)

Compute resources

Wall time 111.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 00:16Z.